A Data-Mining Approach for Investigating Social and Economic Geographical Dynamics of β-Thalassemia's Spread

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***Abstract*—β-Thalassemia is an anemic genetic disorder that remains a major global health issue, especially in the globalized era where public health, economics, and education are tightly interwoven. Previous studies have examined the disease's rate and heredity. This study analyzed β-thalassemia's socioeconomic geography and how it affects the afflicted population. We processed survey data and performed data mining using self-organizing maps to identify underlying data structure. We hypothesized that certain variables mark subgroups within the affected population and we aimed at identifying these subgroups and used a correlation-based measure to assess the variable's importance to the subgroup's distinction. The population's education level was one of the major factors that divided it into different subgroups. Our study showed that recurring patterns of specific variables separated the affected population into disparate subgroups based on their response to questionnaires. Future studies can use such tools to delve deeper into how other variables (e.g. socioeconomic and genomic) can identify subgroups within larger affected populations.**

***Index Terms*—Data mining, β-thalassemia, self-organizing map.**

I. Introduction

In THIS study, we investigated β-thalassemia's socioeconomic geography by combining typical sociological interpretation of extensive data collected via questionnaires with effective exploratory analysis using computational tools such as the self-organizing map (SOM). SOMs are an artificial neural network subtype commonly used for unsupervised analysis (clustering) that produces low-dimensional representation of high-dimensional data. SOMs determine clusters within the data plus other relevant information for further exploitation. They have been successfully used in analyzing multivariate, socioeconomic, and genomic [1]–[3] data. Besides the obvious technical benefits highlighted in the following sections, our choice for the SOM was also motivated by its extensive successful applications in several areas of biomedical engineering [4]–[7]. The visualization benefits and the high-level learning capabilities of the SOM (typical for neural networks-based methods) facilitate our high-dimensional data analysis. As shown by Bonato *et al.*, vector quantization algorithms reduce the feature space's size without losing information for identifying clusters in the classification space [8]. Vector projection algorithms such as multidimensional scaling, Sammon projection can perform projections from a high-dimensional input space into a 3-D output space [8].

β-Thalassemia is an inherited, homozygous recessive genetic disorder that affects the human blood's hemoglobin (Hb) amount. Investigators first coined the first thalassemia in 1925 while studying anemia occurring in children of Italian origin [9]. Successive studies found children of Mediterranean origin affected by this new anemia (from *thalassa*, Greek for sea) [10]. β-Thalassemia continues to affect populations in the Mediterranean, the Maghreb, the Indian Subcontinent, Southwest, South, and Southeast Asia with estimates ranging from 3% to 10% in some areas [11]. β-Thalassemia can be divided into two categories: heterozygous β-thalassemia (minor) produces mild anemia with little or no symptoms. Homozygous β-thalassemia (major) is a severe condition requiring repeated blood transfusions throughout the patient's life [12].

While previous investigations focused on how β-thalassemia spreads within populations, none investigated why it spreads within populations [13], [14]. The inattentiveness to why β-thalassemia spreads has significant potential public health implications.

In this study, we use data-mining analysis methods based on SOM to investigate β-thalassemia's socioeconomic geography in Antalya Province, Turkey since the area has a high number of consanguineous marriages and a high rate of β-thalassemia cases is reported [14], [15]. Our ultimate goal is to expand this study to several different areas in the Mediterranean, South Asia, and other affected areas [12], [13], [16].

II. Methods

A. Data Collection and Preprocessing

We focused on two hospitals, Akdeniz University Hospital and Antalya State Hospital in Antalya Province, in Southwest

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Turkey, because these hospitals are the only hospitals in Turkey that provide specialized services to patients with β-thalassemia. The patients who came from outside Antalya Province stated that doctors from their provinces recommended that they should go to these two hospitals for blood transfusion, iron-chelation treatment, and information.

A questionnaire consisting of 14 questions was used in the study. These questionnaires were created in cooperation with Dr. Akif Yesilipek of Akdeniz University Hospital and Dr. Duran Canatan, head of the Thalassemia Center in Antalya State Hospital. These doctors recommended questions to be used on the questionnaire based on their previous research [14] and research conducted in other studies [17]–[19]. All subjects in the specialized units received a copy of the survey. Our main hypothesis is that certain variables mark subgroups within the affected population. We aimed at identifying these subgroups via clustering analysis. The current study incorporates questions dealing with several aspects of the disease since it is part of a larger study.

The questions asked for the respondent's education, income and job status, β-thalassemia awareness (and where they received the information), β-thalassemia spread awareness, whether β-thalassemia information is readily available in their neighborhoods, whether neighbors were aware of β-thalassemia, which hospitals they visited (and how often), neighborhood prevention measures, treatment affordability and difficulty, whether enough is being done to combat β-thalassemia, and current health insurance. Previous studies used similar variables in selecting questions from their questionnaires when measuring socioeconomic health outcomes [20], [21]. The analysis aims to identify patterns in the respondent's answers that indicate how certain variables' values mark the subgroups in our population.

We created a database consisting of all of the respondent's answers. Data vectors were obtained by digitizing the corresponding respondent answers to the 14 questions. The z-score normalization we subsequently performed prevents bias induced by the range of certain questions in the subsequent analysis [22]. The main problem was mining the high-dimensional data and identifying the factors that induce different characteristic patterns contributing to the subgroups formation within the respondent population.

The dataset is marked by specific characteristics that make data-mining techniques particularly suitable for this analysis. The high-dimensionality and the fact that we are interested in isolating subsets of the data exhibiting similar patterns render the use of SOM highly desirable. Since SOMs are often employed in data mining due to their benefits in term of graphical representation of data and full availability of the output results to further insight and analysis [8]. Thus, an SOM was used to group the data into clusters that brought data vectors together via distance among them [23].

B. Self-Organizing Maps

SOMs are a subtype of artificial neural network commonly used for unsupervised analysis (clustering) that produces low-dimensional representation of high-dimensional data [23]. A SOM is a single-layer network, where one layer of neurons (projecting the input data) represents the output space. Each neuron corresponds to a cluster with an attached weight vector. The weight vectors have the same dimensionality as the input data vectors and their values reflect the content of the cluster they are attached to. The SOM successively presents the available data to the network, with the goal of associating similar data vectors to the same neurons.

The entire training process uses competitive learning by presenting new input data to the network that determines the closest weight vector (using a distance measure, commonly the Euclidian distance) and assigns the data vector to the corresponding neuron. The best-matching neuron's weights and its closest neighbors adapt to reflect the newly assigned data input vector's value. The change's magnitude decreases for neurons physically farther from the best-matching neuron, and during successive time steps. The entire process repeats for all input vectors and during several time epochs. The training process stops once convergence criteria are met [23].

Additionally, the clustering converts data onto a lower dimensional space, determined by the SOM's 2-D structure. The data vector's topology and distance in the input space (of higher dimensionality) are preserved in the output space. This approach allows us to reduce the feature space's size (input space) without losing useful information for the identification of clusters within the population.

The visualization of the algorithm's analysis is essential to explore the data's overall shape and subgroup structure. This is where the SOM outperforms classic statistical methods that are used in the analysis of questionnaire data [24], [25]. Graphical representations, such as the U-matrix, are popular tools that allow the SOM output's visualization and are typically used for demonstrating existing clustering structures. The U-matrix is a grid whose color-coded elements match the distances among the units (neurons) of the SOM's output space. It can be interpreted to indicate the boundaries of clusters in the SOM. But how accurate are these visual representations? We subsequently performed hierarchical clustering on the weight vectors corresponding to the SOM output neurons to obtain a quantifiable answer to this and to identify the cluster-respondent correspondence. This approach has been successfully used to identify larger subgroups within the SOM-clustered data [26].

After identifying the existent subgroups in the data, we proceeded to the final goal of our analysis, which is the determination of the variables responsible for the discrimination between groups. We accomplished this by using a correlation-based measure that is able to assess pairwise differences among subgroups and is similar to the effect size measure commonly used in statistical meta-analysis studies [27]:

$$P_{ij}(v) = \frac{|\mu_i(\mathbf{x}_v) - \mu_j(\mathbf{x}_v)|}{\sigma_{\text{pooled}}} \tag{1}$$

where $\mu_i(\mathbf{x}_v)$ and $\mu_j(\mathbf{x}_v)$ denote the mean values for the variable v in group i and j, respectively (i.e., the respondents' answers normalized values to the vth question in the two groups); σ_{pooled} is the root mean square of the two standard deviations [27]. A large $P_{ij}(v)$ indicates strong correlation between

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the variable v and the group distinction, and thus the measure is used to identify "strong" variables.

In the last step of our analysis, we investigate whether applying a different clustering approach yields similar results (in terms of clustering membership) with those obtained via our analysis. For that we employ a method based on multiple correspondence analysis (MCA) [28]. MCA is a classical statistical method that has been successfully used before in identifying heterogeneous subgroups of respondents from questionnaire's data [29], [30].

III. Results

We based our map size ($9 \times 7 = 63$ nodes for the SOM) on Vesanto and Kaski's criteria [24], [25], [31]. Namely, the formula $5 \times \sqrt{\text{number of samples } n}$ determines the number of map neurons. The map size is computed in the following manner: the data's two largest eigenvalues are calculated and the ratio between the column and row numbers of the SOM map grid was set to the ratio between the two largest eigenvalues. The final columns and rows were then set so that their product closely paralleled the number of map nodes stated earlier. We chose this method to ensure fairly balanced clustering [23].

Our database consisted of $n = 142$ data vectors (corresponding to the number of respondents), each containing the answers to $m = 14$ questions, resulting in a 142×14 data matrix. We screened for, and eliminated, surveys that contained unanswered questions before the start of our analysis. After normalizing the data, we developed and trained an SOM ($9 \times 7 = 63$ nodes) using our data matrix. The SOM's topology was chosen to be a 2-D sheet containing a hexagonal lattice of map units (neurons). The SOM was trained using a batch-training scheme for 1000 epochs [25], [31] using a Gaussian-type neighborhood function and a linear initialization scheme. Our unsupervised analysis clustering identified respondent groups based on their response patterns to the questionnaire's questions.

The resulting SOM can be visualized in numerous ways, one of them being that of component planes, as shown in Fig. 1. Each component plane displays the distribution across SOM neurons of the values for a single variable of the data space, which are color coded according to their range. Please note that for practical reasons Fig. 2 presents only 10 of the 14 component planes corresponding to the 14 data variables. The 10 component planes displayed correspond to the variables identified in Table II as relevant for subgroup discrimination, as described later.

The respondents with high values of the variable "Education" seem to be grouped in the center-left part of the map. Conversely, the lowest values of the variables "Thalassemia Awareness," "Spread Awareness," "Info. Availability," and Neighborhood Awareness," are located at the center-right side of the map.

As noted earlier, to gain further insight into the data-clustering structure. We are using a unified distance matrix (U-matrix), which enhances structure visualization within the SOM. The U-matrix can be interpreted to indicate the SOM cluster's boundaries, which match the region's color-coded values toward the upper side of the scale in the Fig. 1, contrasted to regions coded with colors toward the scale's lower side [23].

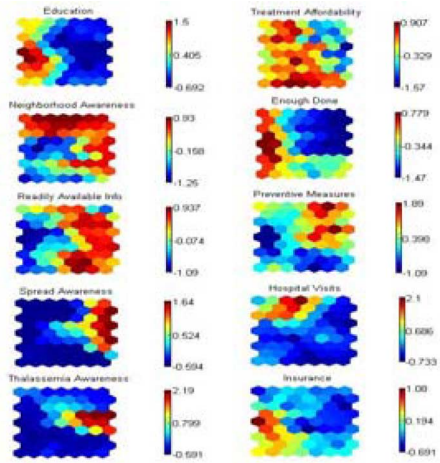


Fig. 1. Colormaps representing 10 of the 14 variables that correspond to the questionnaire's 14 questions. Each subplot presents the values of its corresponding variable over all the map nodes. Scales to the right of each subplot represent values of each variable. Dark red correspond to the highest values, while dark blue the lowest.

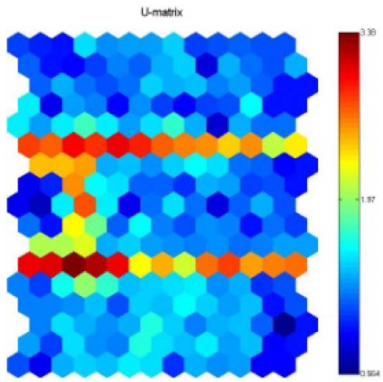


Fig. 2. U-matrix. Note the four distinct regions representing the four groups. Relative distances among adjacent clusters are higher in the areas separating the groups, marked in the plot by color tones toward red.

This is particularly helpful in the unsupervised analysis scenario, when the user has no a prior information on the number of groups within the data, which is also the case with our analysis. Fig. 2 depicts the U-matrix and offers a fast way to get insight into the data distribution.

The U-matrix represents another image of our SOM map. There are four distinct regions separating the entire SOM. Three are parallel to each other while the fourth region is perpendicular to the three parallel regions and is facing left. The four regions represent the four groups in this study. We performed an agglomerative hierarchical clustering on the representative vectors resulting from the SOM training [26] to obtain an accurate estimate of the group's structure and distance among them. In the resulting dendrogram (Fig. 3), the four groups visually identifiable from the U-matrix in Fig. 2 were easily retrieved when setting a cut at value 2 on the distance (vertical) axis.

The dendrogram's x-axis represents the SOM's weight vectors and the y-axis represents the weight vector's distances. The dendrogram takes each data vector closest to each other and clusters them together. You have corresponding distance in the

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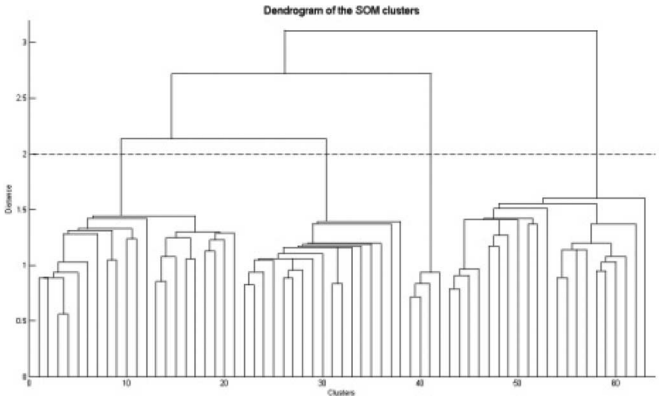


Fig. 3. SOM dendrogram. This structure allowed us to view the cluster's hierarchical representation, structure, and relative distance, and thus to identify the final group membership of each SOM cluster.

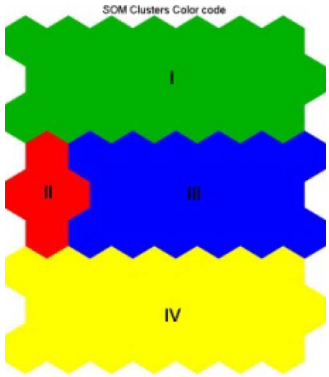


Fig. 4. Final SOM cluster structure after superimposing the results of the hierarchical clustering of the SOM.

y-axis: The longer the vertical line when two data vectors are clustered together, the further they separate. Upon overlaying the dendrogram and the U-matrix, we identified the four groups as highlighted in Fig. 4.

We further scrutinized the groups' separation and the variables responsible for this to gain more information on the data's partitioning, based on the measure defined in (1). The measure reflects the pairwise differences between the four groups and indicates variables more relevant for the group separation. Table I presents the ten variables that rank as top five separating variables within the groups (highest five $P_{ij}(v)$ values marked in bold) for each pairwise comparison between groups. Variable "Education" is the top separator in two of the six pairs. The other variables, "Neighbor Awareness, Enough Done, Treatment Affordability," and "Awareness" (of what β-Thalassemia is), are the prime factors in the other four pairs, one per pair.

A previous study highlighted how highly educated individuals were less likely to pursue consanguineous marriages [15]. Group II was the only group among the four with advanced degrees. This group (compared to others) contained respondents with higher salaries than other groups that had no respondents with advanced degrees. Despite most of group II claiming that they can afford treatment (with difficulties), more were able to fully afford treatment than any other group.

TABLE I
PAIRWISE INDICATORS OF VARIABLE RELEVANCE IN THE GROUPS SEPARATION

P(v) - among groups	***G1/G2***	***G1/G3***	***G1/G4***	***G2/G3***	***G2/G4***	***G3/G4***
Education	**1.28**	**1.06**	0.70	**1.73**	**1.11**	**1.26**
Neighborbood Awareness	**1.63**	**0.64**	**0.87**	**1.51**	0.32	0.29
Readily Available Info	**1.41**	0.21	0.31	0.60	**0.93**	0.18
Spread Awareness	**0.76**	0.28	0.32	**0.94**	0.10	**0.95**
Thalassemia Awareness	0.26	**1.04**	**0.77**	0.76	0.16	**1.40**
Treatment Affordability	0.21	**0.55**	0.53	0.22	0.46	**0.88**
Enough Being Done	0.47	0.23	**1.33**	**1.25**	**0.71**	**0.97**
Preventive Measures	0.60	0.29	**1.08**	**0.79**	**0.82**	0.51
Hospital Visits	0.34	**0.74**	**0.81**	0.42	0.38	0.73
Insurance	**0.72**	0.11	0.62	0.61	**0.63**	0.63

Later, we present a breakdown of the characteristics of the four subgroups and the responding answering patterns of the respondents identified through our analysis as members of the corresponding subgroups. Note that the variables (questions) described match the ten variables emerged as significant for the subgroup distinction.

Group I has the second highest number of respondents who had no education (after group III). Our unsupervised clustering placed together respondents that are roughly split on what β-thalassemia is and how it spreads. Over 60% are unaware of readily available information in their neighborhoods and 90.4% do not know if their neighbors know of this disease. An overwhelming majority do not know if their neighbors are aware of this disease or of any preventative measures taken by their neighborhoods. A slight majority, 53.8%, believe enough is being done to combat this disease. Most of these respondents (67.3%) visit the hospital less than once a month. This group has the second lowest average income (after group III).

Group II is the only group containing respondents with advanced education (university and masters') and has the highest average income of the four groups. Group II is overwhelmingly aware of β-thalassemia, how it spreads, readily available information, and preventative measures in their neighborhoods. The majority feels that little is being done overall to combat this disease. The respondents visit the hospital more often than any other group.

Group III has the highest percentage of respondents without education. The majority does not know what β-thalassemia is and how it spreads, cannot identify readily-available information and preventative measures in their neighborhoods, and feels that enough is being done to combat this disease. Over half are aware that their neighbors know of β-thalassemia. Over 60% are unable to afford treatment. This group has the lowest monthly income of the four groups.

Group IV has the second highest education level of the four groups (after group II). The majority is aware of what

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TABLE II
BREAKDOWN OF THE FOUR SUBGROUPS AND RESPONDENT'S ANSWER PATTERNS

	Group I	Group II	Group III	Group IV
Number of Respondents	52	12	36	42
Education (%)				
- **Primary**	48	16.7	66.7	33.3
- **Secondary**	13.6	0	11.1	16.7
- **First Tier-High School**	30.7	25	5.5	26.2
- **Second Tier-High School**	0	16.7	0	23.8
- **University**	0	33.3	0	0
- **Masters**	0	8.3	0	0
Health Insurance (%)				
- **Emekli**	5.76	41.67	0	11.9
- **Bagkur**	34.63	33.33	16.67	35.7
- **SSK**	42.31	25	58.33	35.7
- **Green Card**	15.38	0	19.45	16.7
- **None**	1.92	0	5.55	0
What β-thalassemia is (%)				
- **Yes**	52	100	33.3	83.3
- **No**	48	0	66.7	16.7
How β-thalassemia spreads (%)				
- **Yes**	66.7	36.2	91.6	46.1
- **No**	33.3	63.8	8.6	53.9
Readily available information (%)				
- **Yes**	38.5	83.3	22.3	50
- **No**	61.5	16.7	77.7	50
Neighbor Awareness (%)				
- **Yes**	9.6	75	55.5	52.4
- **No**	90.4	25	44.5	47.6
Preventative Measures (%)				
- **Yes**	13.4	83.3	25	52.4
- **No**	86.6	16.7	75	47.6
Is enough being done (%)				
- **Yes**	46.2	8.6	66.6	64.3
- **No**	53.8	91.6	33.4	35.7
Hospital Visits (%)				
- **Less than Monthly**	67.3	16.7	58.3	42.9
- **Monthly**	28.8	50	22.2	33.3
- **More than Monthly**	3.9	33.3	19.5	23.8
Treatment Affordability (%)				
- **Yes**	17.3	25	5.6	26.2
- **Yes + Difficulties**	38.4	58.3	33.3	33.3
- **No**	44.3	16.7	61.1	40.5
Average Monthly Income (USD)	516.75	751.67	463.77	577.88

β-thalassemia is and how it spreads and is roughly split on neighbor awareness and neighborhood preventative measures. Most (64.3%) feel that little is being done to rout this disease. Over 40% visit the hospital less than once a month and over 40% are unable to afford treatment.

We checked the respondent's health insurance coverage because Turkey's health insurance system combines public and private health insurance plans [32]. These plans vary according to each respondent's current economic status. We noted that higher income leads to increased insurance coverage and treatment. According to Table II, group II has the highest concentration of respondents with advanced education. Furthermore, group II has the highest monthly income. Social Insurance Association (SSK) is for private and blue-collar sector public sector workers. Emekli is a government-fund for retired civil servants. Bagkur is for self-employed persons, and the Green Card is for persons receiving below minimum wages.

In the last step of our analysis, we have applied a method based on MCA [28] to further validate our results. MCA has been successfully used before in identifying heterogeneous respondent subgroups from questionnaires categorical data [29], [30]. Since MCA is a factor-analysis-based method that represents variables in a property space (mapping variables' associations into fewer dimensions), we have used it within the same framework as in [29]. Namely, we employed MCA for finding a low-dimensional space representation of the original multivariate categorical variables from our questionnaires. Simultaneously, a k-means clustering scheme was used to identify clusters of respondents within the data that are relatively homogeneous in terms of the new low-dimensional representation. This is achieved by minimizing a composite objective function described in [29].

For the MCA factorization, the number of dimensions used for the new data representation was $d = 3$ (which explains 61.8% of the total inertia), based on the $1/J$ criterion described in [7] (where J is the total number of variables in the data). Interestingly, the optimal number of clusters that minimized the objective function was $c = 4$. When comparing the members of the clusters obtained via the MCA and k-means clustering method to the members of the groups we have obtained from our SOM and hierarchical clustering method, we found that: 88.5% (46 out of 52) of the respondents in group I, 100% (12 out of 12) of the respondents in group II, 88.9% (32 out of 36) of the respondents in group III, and 90.5% (38 of 42) of the respondents in group IV, were clustered together as in their original groups. It must be noted that the four clusters obtained from MCA and k-means had 48, 12, 37, and 45 members, respectively. This consistency in grouping the respondents is highly indicative of the existence of certain structures/trends within the data determined by the dominant variables presented in Table I, and of the relevance of these structures to our analysis.

IV. DISCUSSION

This study analyzed β-thalassemia's socioeconomic geography and how it affects the afflicted population. We gathered information on β-thalassemia's social and economic geography using interviews and surveys, and combined this method with the SOM.

The survey questions and the population's response to them turned the population into four subgroups. We noted correlations in each subgroup's answers to the surveys, particularly in "Education," "Awareness," "Spread Awareness," and "Enough Done" in Fig. 1.The variables in Fig. 1, when fused with the U-matrix and dendrogram, portray a disease affecting different societal levels differently. It appeared that respondents within group II (which has the highest education level among the four groups) were much more aware of what β-thalassemia is, how it spreads, and felt that more must be done to rout this disease. It is also worth noting that, according to Fig. 2, group III (which has the lowest education level among the four groups) are less aware of what β-thalassemia is, how it spreads, and (mostly) feel that enough is being done.

According to Table I, group II has the highest concentration of respondents with advanced education. Furthermore, group II has the highest monthly income. SSK is for private and blue-collar sector public sector workers. Emekli is a government-fund for retired civil servants. Bagkur is for self-employed persons, and the Green Card is for persons receiving below minimum wages.

We noted that although only group II contained respondents with advanced education, but had the lowest number

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covered under SSK, they made more per month than the other respondents in other group. We further noted that nobody in group II was covered under the Green Card. Despite respondents in group II possessing advanced degrees, most of this group has difficulty paying for treatment, although they could afford the treatment. We speculated that the reason for group II's difficulty paying for treatments may reflect on the group's critical view of the entire situation and how their education may have given them another perspective on the disease. Respondents in groups I, III, and IV (44.3%, 61.1%, and 40.5%, respectively) were unable to afford treatment. Multiple factors may be responsible: transportation costs (driving long distances for iron-chelation treatment unavailable elsewhere, which was also a major cost factor for other groups), absence from work (meant no pay), and finding a hotel (for family members) to stay while receiving treatment, and because patients must still pay a fee for drugs and medication.

Although the study's time frame was limited (two and a half months), future studies can delve deeper into how social and economic policies, and genetics can predict β-thalassemia's spread. Future studies could look into other countries to see if similar patterns exist in regions affected by β-thalassemia and how the disease's social, economic, and genetic geography can predict its spread and how targeted prevention policies can slow down or halt this disease's spread.

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